



5.4.4 Proof of the Cholesky Factorization Theorem



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fit width

Partition, once again,

$$A \rightarrow \begin{pmatrix} \alpha_{11} & a_{21}^H \\ a_{21} & A_{22} \end{pmatrix}.$$

The following lemmas are key to the proof of the Cholesky Factorization Theorem:

Lemma 5.4.4.1. *Let $A \in \mathbb{C}^{n \times n}$ be HPD. Then α_{11} is real and positive.*

Proof.

This is special case of [Homework 5.4.1.2](#).

Lemma 5.4.4.2. *Let $A \in \mathbb{C}^{n \times n}$ be HPD and $l_{21} = a_{21} / \sqrt{\alpha_{11}}$. Then $A_{22} - l_{21} l_{21}^H$ is HPD.*

Proof.

Since A is Hermitian so are A_{22} and $A_{22} - l_{21} l_{21}^H$.

Let $x_2 \in \mathbb{C}^{n-1}$ be an arbitrary nonzero vector. Define $x = \begin{pmatrix} \chi_1 \\ x_2 \end{pmatrix}$ where $\chi_1 = -a_{21}^H x_2 / \alpha_{11}$. Then, since clearly $x \neq 0$,

$$\begin{aligned} &0 \\ &< \quad < A \text{ is HPD} > \\ &x^H A x \\ &= < \text{partition} > \\ &\begin{pmatrix} \chi_1 \\ x_2 \end{pmatrix}^H \begin{pmatrix} \alpha_{11} & a_{21}^H \\ a_{21} & A_{22} \end{pmatrix} \begin{pmatrix} \chi_1 \\ x_2 \end{pmatrix} \\ &= < \text{partitioned multiplication} > \\ &\begin{pmatrix} \chi_1 \\ x_2 \end{pmatrix}^H \begin{pmatrix} \alpha_{11} \chi_1 + a_{21}^H x_2 \\ a_{21} \chi_1 + A_{22} x_2 \end{pmatrix} \\ &= < \text{partitioned multiplication} > \\ &\alpha_{11} \bar{\chi}_1 \chi_1 + \bar{\chi}_1 a_{21}^H x_2 + x_2^H a_{21} \chi_1 + x_2^H A_{22} x_2 \\ &= < \text{linear algebra} > \\ &\alpha_{11} \frac{a_{21}^H x_2}{\alpha_{11}} \frac{x_2^H a_{21}}{\alpha_{11}} - \frac{x_2^H a_{21}}{\alpha_{11}} a_{21}^H x_2 - x_2^H a_{21} \frac{a_{21}^H x_2}{\alpha_{11}} + x_2^H A_{22} x_2 \\ &= < \text{since } x_2^H a_{21}, a_{21}^H x_2 \text{ are scalars and hence can move around; } \alpha_{11} / \alpha_{11} = 1 > \\ &x_2^H a_{21} \frac{a_{21}^H x_2}{\alpha_{11}} - x_2^H a_{21} \frac{a_{21}^H x_2}{\alpha_{11}} - x_2^H a_{21} \frac{a_{21}^H x_2}{\alpha_{11}} + x_2^H A_{22} x_2 \\ &= < \text{cancel terms; factor out } x_2^H \text{ and } x_2 > \\ &x_2^H \left(A_{22} - \frac{a_{21} a_{21}^H}{\alpha_{11}} \right) x_2 \\ &= < \text{simplify} > \\ &x_2^H (A_{22} - l_{21} l_{21}^H) x_2. \end{aligned}$$

We conclude that $A_{22} - l_{21} l_{21}^H$ is HPD.

Proof of the Cholesky Factorization Theorem.

Proof by induction.

1. Base case: $n = 1$:

Clearly the result is true for a 1×1 matrix $A = \alpha_{11}$: In this case, the fact that A is HPD means that α_{11} is real and positive and a Cholesky factor is then given by $\lambda_{11} = \sqrt{\alpha_{11}}$, with uniqueness if we insist that λ_{11} is positive.

2. Inductive step: Assume the result is true for $n = k$. We will show that it holds for $n = k + 1$.

Let $A \in \mathbb{C}^{(k+1) \times (k+1)}$ be HPD. Partition

$$A = \begin{pmatrix} \alpha_{11} & a_{21}^H \\ a_{21} & A_{22} \end{pmatrix} \text{ and } L = \begin{pmatrix} \lambda_{11} & 0 \\ l_{21} & L_{22} \end{pmatrix}.$$

Let

- $\lambda_{11} = \sqrt{\alpha_{11}}$ (which is well-defined by [Lemma 5.4.4.1](#),
- $l_{21} = a_{21}/\lambda_{11}$,
- $A_{22} - l_{21}l_{21}^H = L_{22}L_{22}^H$ (which exists as a consequence of the Inductive Hypothesis and [Lemma 5.4.4.2](#).)

Then L is the desired Cholesky factor of A .

3. By the Principal of Mathematical Induction, the theorem holds.